

Macroeconomics Notes

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1 Gross Domestic Product (GDP)

Gross Domestic Product (GDP) is the total market value of all **final goods and services** produced **within a country's borders** in a **specific time period**.

GDP does not include:

- Intermediate goods
- Non-market activities
- Used goods transactions
- Financial transactions
- Illegal activities
- Environmental externalities

2 Methods of Measuring GDP

1. Production (Value-Added) Approach:

- Aggregates the value added at each stage of production.
- Value added = Revenue – Cost of intermediate inputs.

2. Expenditure Approach:

- Adds up total spending on final goods and services.
- $GDP = \text{Consumption} + \text{Investment} + \text{Government Spending} + (\text{Exports} - \text{Imports})$.

3. Income Approach:

- Sums all incomes earned by households and businesses during production.
- $GDP = \text{Labor income} + \text{Capital income}$.
- $GDP = \text{Wages} + \text{Rent} + \text{Interest} + \text{Profit} + \dots$

3 Gross National Product (GNP)

Gross National Product (GNP) is the total market value of all **final goods and services** produced by a country's residents and businesses, regardless of where that production takes place during a specific time period.

- GDP focuses on *where* production happens.
- GNP focuses on *who* produces it (the country's citizens and businesses).

The relationship between GNP and GDP:

$$\text{GNP} = \text{GDP} + \text{Net Factor Income from Abroad}$$

Where "Net Factor Income from Abroad" represents income earned by domestic residents from foreign investments minus income earned by foreign residents from domestic investments. This means if a country earns more from abroad than foreigners earn domestically, GNP will be higher than GDP, and vice versa.

4 National Income Accounting Identity

The relationship between a country's total income and its expenditures:

$$Y = C + I + G + (X - M)$$

Where Y is GDP, C is household consumption, I is business investment, G is government purchases, and $(X - M)$ is net exports.

1. Consumption (C)

- *Definition:* All household spending on goods and services, excluding new housing.
- *Includes:*
 - **Durable Goods:** Long-lasting items (cars, household appliances, furniture).
 - **Non-Durable Goods:** Quick-use items (food, clothing, gasoline).
 - **Services:** Non-physical, intangible products (healthcare, legal services, education).

2. Investment (I)

- *Definition:* Private sector spending on capital goods and changes in inventories.
- *Includes:*
 - **Business Fixed Investment:** Capital expenditures by firms (equipment, factories).
 - **Residential Fixed Investment:** Spending on new housing units by households and landlords (new home constructions and purchases, major home renovations).
 - **Inventory Investment:** Changes in the stock of unsold goods held by firms.
- *Excludes:* Financial assets like stocks, bonds, or cryptocurrencies, as these do not represent production.

3. Government Spending (G)

- *Definition:* Government purchases of goods and services.
- *Includes:*
 - Federal, state, and local spending on goods and services.

- *Excludes:* Transfer payments (social security, unemployment benefits, welfare), because these are not payments for goods or services.

4. Net Exports ($\text{NX} = \text{Exports} - \text{Imports}$)

- *Definition:* The difference between exports (X) and imports (M).
- *Includes:*
 - **Exports (X):** Goods and services produced domestically and sold to foreign buyers.
 - **Imports (M):** Goods and services produced abroad and purchased by domestic consumers.
- Can result in trade surplus (positive) or deficit (negative).

5 Typical GDP Component Distribution

- Consumption (C): $\sim 70\%$ – *Largest component*
- Investment (I): $\sim 15\text{-}20\%$
- Government (G): $\sim 15\text{-}20\%$
- Net Exports (NX): $\sim \pm 5\%$ – *Smallest component*

6 Disposable Personal Income

Disposable Personal Income (DPI) is the amount of money that households have available to spend or save after paying personal income taxes.

$$Y^d = Y - T \quad \text{or} \quad Y^d = C + S$$

Where Y^d is disposable income, Y is total income, T is taxes, C is consumption, S is savings.